

1. Describe the main difference between a gated S-R latch and an edge-triggered S-R flip-flop.

The main difference between a gated S-R latch and an edge-triggered S-R flip-flop lies in how they respond to their control signals. A gated S-R latch is level-sensitive, meaning its output can change as long as the enable signal is active. An edge-triggered S-R flip-flop is edge-sensitive, so it changes its output only on a specific clock transition, either the rising or falling edge.

2. How does a JK flip-flop differ from an SR flip-flop in its basic operation?

A JK flip-flop differs from an SR flip-flop in its handling of inputs. While an SR flip-flop has an invalid state when both S and R are high simultaneously, the JK flip-flop eliminates this problem by toggling the output when both J and K inputs are high.

3. Discuss the application of flip-flops in data storage.

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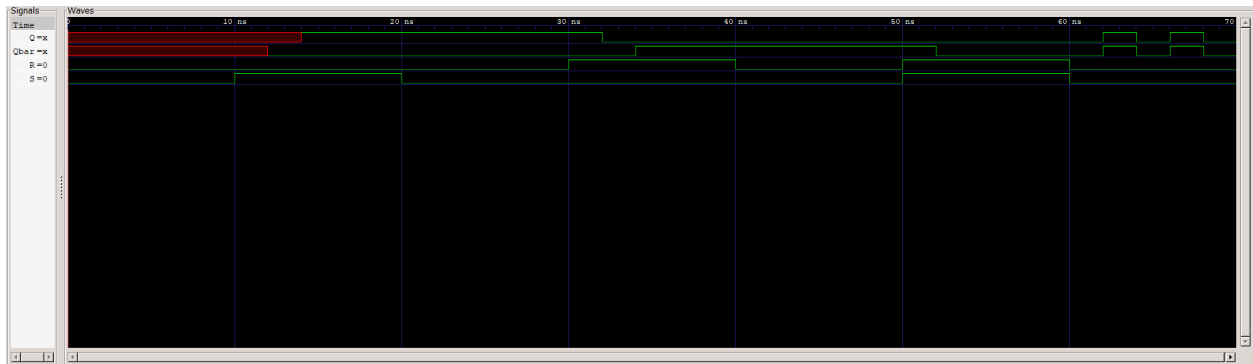
4. What are synchronous and asynchronous circuits?

Synchronous circuits rely on a common clock signal where state changes occur at defined clock edges. Asynchronous circuits do not depend on a clock signal and their outputs change immediately in response to input changes.

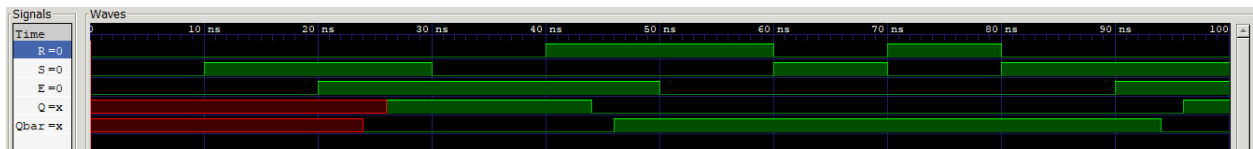
5. How many flip-flops do you require storing the data 1101?

Four flip-flops are required, since each flip-flop stores exactly one bit of information.

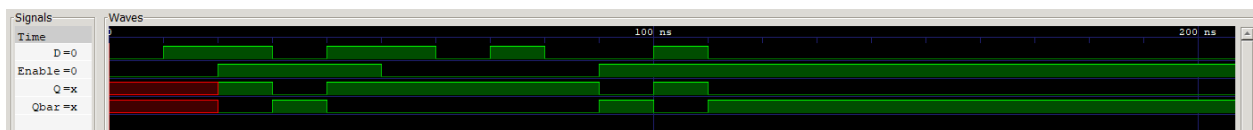
Task 1



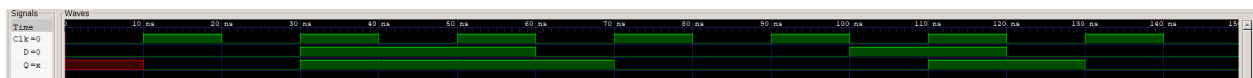
Task 2



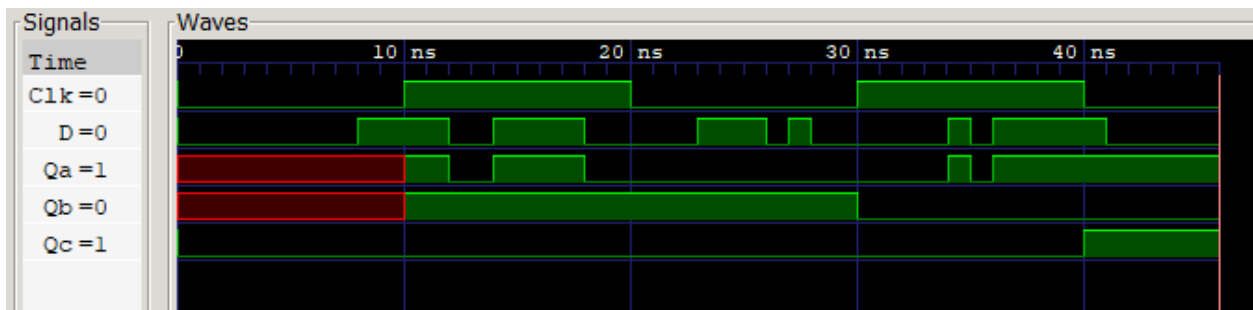
Task 3



Task 4



Task 5



Task 6

